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title: "BDA400 Assignment 5 – Technical Analysis using R"

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Introduction

This assignment implements and tests technical analysis indicators in R. The functions created include SMA, EMA, MACD, standard deviation, linear regression, RSI, Stochastic RSI, crossover, and crossunder.

SMA (Simple Moving Average)

The SMA function calculates the average of the most recent values over a specified period.

The function was tested using the following data:

```
sma(c(10, 12, 15, 20, 18, 22, 25, 24, 21), 3)
~~~
## EMA (Exponential Moving Average)

The EMA function calculates a weighted average that gives more importance to recent values.

```r
ema(c(10, 12, 15, 20, 18, 22, 25, 24, 21), 3)
```

MACD (Moving Average Convergence Divergence)

The MACD function calculates the difference between short-term and long-term moving averages.

```
macd(c(10, 12, 15, 20, 18, 22, 25, 24, 21), 3, 5, 2)
```

MACD (Moving Average Convergence Divergence)

The MACD function calculates the difference between short-term and long-term moving averages.

```
macd(c(10, 12, 15, 20, 18, 22, 25, 24, 21), 3, 5, 2)
```r

## Standard Deviation

The standard deviation function measures the amount of variation in the data.

```r
stdev(c(10, 12, 15, 20, 18, 22, 25, 24, 21))
```r

## Linear Regression

The linear regression function calculates the slope, intercept, and predicted values.
```

```r
linreg(c(10, 12, 15, 20, 18, 22, 25, 24, 21), 5, 0)
```

Relative Strength Index (RSI)

The RSI function measures the speed and change of price movements and helps identify overbought or oversold conditions.

```r
rsi(c(10, 12, 15, 13, 16, 18, 17, 20, 22, 21), 3)

```r
## Stochastic RSI
The Stochastic RSI measures the position of the RSI relative to its recent range and helps identify ove
```r
stoch_rsi(c(10, 12, 15, 20, 18, 22, 25, 24, 21), 3, 3, 3)
```r
## Crossover

The Crossover function identifies when one indicator moves above another indicator and can be used to i

```r
crossover <- function(arr1, arr2) {
    if (length(arr1) != length(arr2)) {
    stop("Both arrays should have the same length")
  }
    result <- rep(FALSE, length(arr1))
    for (i in 2:length(arr1)) {
          if (arr1[i] > arr2[i] && arr1[i - 1] <= arr2[i - 1]) {
                  result[i] <- TRUE
                        }

  }
   return(result)

}
```

```r
source("crossover.R")
crossover(
  c(10, 12, 15, 20, 18, 22, 25, 24, 21),
  c(11, 14, 14, 19, 19, 21, 23, 25, 20)
)
```

```
## [1] "None" "None" "Up"   "None" "Down" "Up"   "None" "Down" "Up"
```

Crossunder

The Crossunder function identifies when one indicator moves below another indicator and can be used to identify potential trend changes.

```r
source("crossunder.R")
crossunder(
  c(18, 20, 22, 18, 15, 12, 10, 11, 13),
  c(10, 12, 15, 20, 18, 22, 25, 24, 21)
)
```

```
## [1] "None" "None" "None" "Down" "None" "None" "None" "None" "None"
```